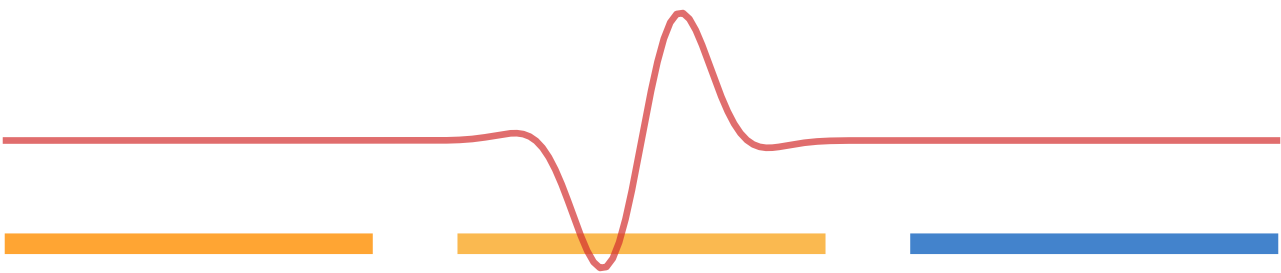


Jing-Zhang Pulse

China's first self-designed railway reawakens as an urban heartbeat in the AI era



TheKireal

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概念方案 | Conceptual proposal
所有内容均为概念建议，可供专业团队深化研究

1. Core Concept

Jing-Zhang Pulse

The Jing-Zhang Railway was China's first self-designed railway. A century later, high-speed rail went underground and the old surface line became a 9 km green corridor. Line 13 trains still shuttle past on the elevated viaduct — every 3-5 minutes, like a heartbeat.

Jing-Zhang Pulse makes this old line, cut by roads, into a breathing public spine once more. It breathes even when no one is there; when people come, they hear the pulse more clearly.

Three Segments

- South Home (Quiet): Beijing North Station — Zhichunlu, tight against residential areas. Light, calm, no disturbance. AI like an unobtrusive lamp.
- Middle Field (Gather): Zhichunlu — Qinghua East Road, widest section, closest to Line 13. Grass theater, heartbeat point.
- North Line (Flow): Qinghua East Road — North 5th Ring, narrow and natural. Light, fog, sound flow; no gathering. Protect the children's field.

Structure: One Spine, Three Segments, Nine Nodes

Breathing Gate, Old Track Memory, Home Plaza, Heartbeat Point, Grass Theater, Pulse Marker Tower, Mist-Light Corridor, Children's Field, Qinghe Wind Corridor

2. Three Key Interventions

1. Subway Sound-Light Heartbeat Point (Middle)

Where Line 13 viaduct is closest to the park.

- Lights brighten as train approaches
- Light band follows as train passes
- Lights dim in breathing rhythm after train leaves
- No loudspeakers; low-frequency vibration and ambient sound only

2. Grass Theater Human-AI Co-performance Field (Middle)

Sunken grass-slope amphitheater in the middle open lawn.

- Grass for sitting/lying by day
- Step lights create a stage at night
- People can sit, stay, make things happen
- Student concerts, developer demos, community cinema, AI art
- No fixed large equipment; everyday lawn character preserved

3. Pillar Light Cascade (North, Line 13 viaduct pillars)

Line 13 viaduct passes through the park in the north; the pillar row is itself a line.

- Four-sided projection mapping, lighting pillar by pillar as trains pass
- Century of rolling stock: steam locomotives, green-skinned trains, high-speed EMU
- Locomotives face the same direction as the train, past to future
- Warm golden light as art and path illumination; dims after 21:30
- No new structures, no pillar modification, no advertising

4. Old Track Walk Pulse Line (South, full route)

2.4 km of restored 1909 tracks in the south.

- Cultural anchor; path from home to stage
- Walkable, pausable, audible
- Not fenced off or enshrined
- Whisper-level audio posts along the tracks

5. South-Middle Path Gap and Sidewalk Wayfinding

The south and middle sections are separated by buildings; the park path does not connect.

- No demolition or construction; paint tricolor pulse line + arrows on detour sidewalks
- Turns the gap into a ceremonial gate crossing
- Photoluminescent or low-voltage self-illuminating at night; low cost, iterable

3. Scenario Cards (13)

- 01 Heartbeat Train: in-ground lights sync with passing Line 13 trains
- 02 Grass Theater: lawn by day, stage by night, spontaneous performances
- 03 Old Track Whisper: scan QR for railway history and AI soundscapes, whisper level
- 04 Breathing Gate: south entrance, warm white breathing light, welcoming home
- 05 River of Light in Mist: north section fiber optics light with footsteps
- 06 Children's Sound-Painting Wall: voices become colored ripples, gentle
- 07 Developer Demo Corner: semi-outdoor prototype demos and peer exchange
- 08 Qinghe Wind Whispers: wind devices transform river wind into light and sound
- 09 Community Cinema Night: 2-4 times/month, directional speakers
- 10 Pulse Morning Run: foot lights pulse with running rhythm, tricolor gradient
- 11 AI Railway Stories: keyword-generated stories on e-ink displays
- 12 Pulse Data Wall: real-time park vitals, anonymous data only
- 13 Pillar Light Cascade: north viaduct pillars project century of rolling stock

Three AI Test Scenarios

- 1. Train rhythm sensing and light sync (target latency <200ms)
- 2. Directional sound field boundary (55dB at 50m)
- 3. Child-friendly light/sound thresholds (65dB max, no direct eye brightness)

4. Personas, Events & Narrative

Five User Personas

1. South residents: daily life, parenting, downstairs-and-there. Need quiet, safety.
2. AI developers: work, exchange, light demo. Active mainly in the middle.
3. University students: inspiration, socializing, events. Primary theater users.
4. International visitors: recognition, check-in, sharing. Marker tower and heartbeat.
5. Children and parents: safety, fun, inspiration. South and north children's field.

Annual Events

- Spring: Pulse Opening Run (9 km) + Grass Theater Opening Season
- Summer: Community Cinema Night (2-4/month) + Developer Demo Day (monthly)
- Autumn: Jing-Zhang Railway Culture Week + AI Art Light Festival (Pillar Cascade)
- Winter: Warm Light Plan (south warm white + hot drink stations, undisturbing)
- Year-round: Pulse Morning Run (daily early morning, lights pulse with rhythm)
- No large events in the south; middle soundscape $\leq 55\text{dB}$ at 50m

Three AI Pilgrimage Landmarks

1. Heartbeat Point (middle): the ritual of waiting for a train
2. Pulse Marker Tower (middle): vantage point to see the 9 km spine
3. Pillar Light Cascade (north): cinematic sequence from past to future

Cultural Narrative & Guide Route

Narrative: from self-designed railway to self-designed intelligence.

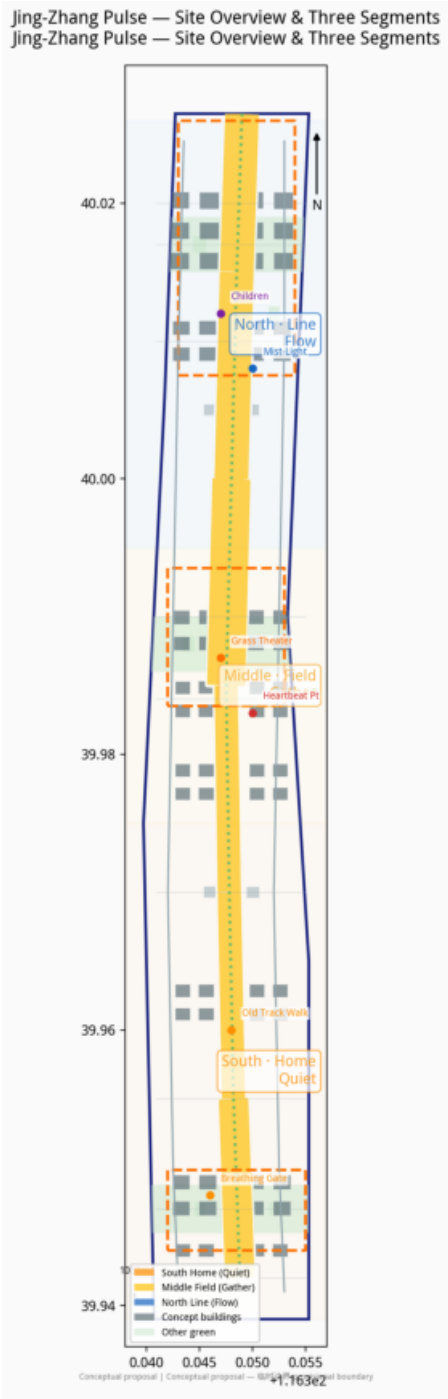
Route (10 stops, ~5 km): Breathing Gate -> Old Track Memory -> Home Plaza

-> Zhichunlu gap gate crossing (sidewalk tricolor detour) -> Heartbeat Point

-> Grass Theater -> Marker Tower -> Mist-Light Corridor -> Pillar Cascade -> Children's Field

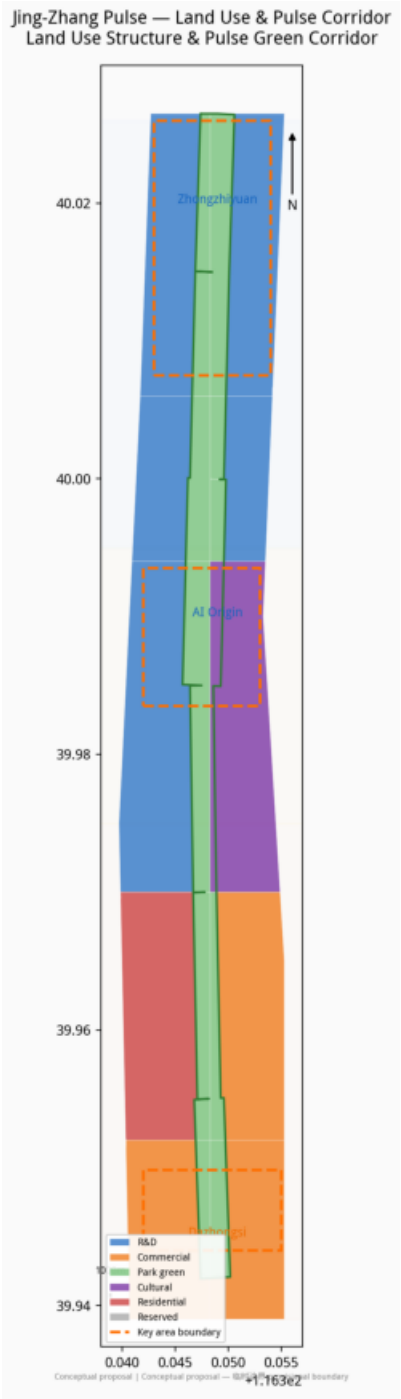
Principles: light intervention, iterable; south residents have veto; quiet hours 22:00-7:00.

Fig 1: Site Overview & Three Segments



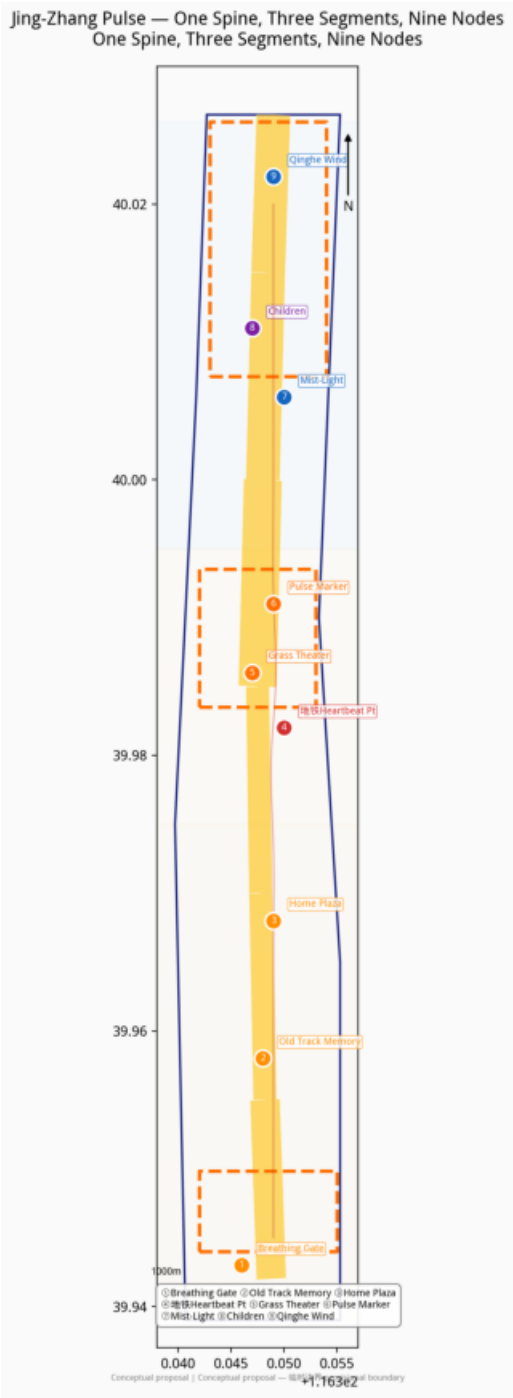
South quiet, middle gather, north flow

Fig 2: Land Use & Pulse Corridor



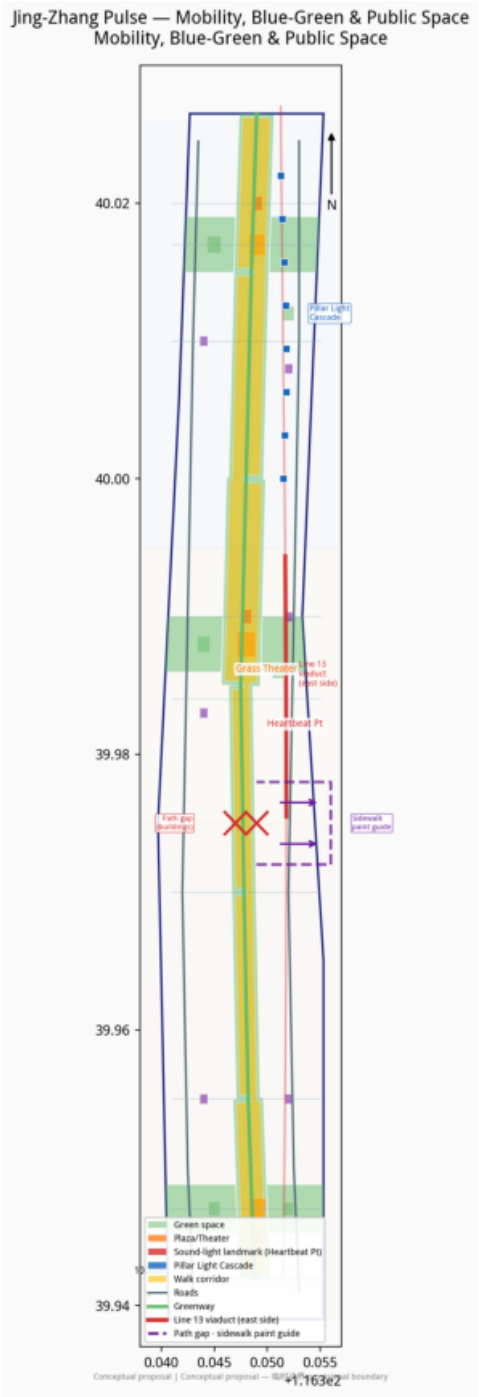
R&D 37.2%, Commercial 22.7%, Park 18.6%

Fig 3: One Spine, Three Segments, Nine Nodes



Nine public space nodes along 9 km

Fig 4: Mobility, Blue-Green & Public Space



Three-path slow system + Line 13 heartbeat

Fig 5: Metrics & Evidence

Jing-Zhang Pulse — Metrics & Evidence | Metrics & Evidence

11.41 km²

Overall scope

11,412,825 m²

29.7%

Green ratio

3,387,189 m²

15.6%

Public space

1,781,980 m²

7.6%

7.7%

≈9 km

南段·家段

守静

N. Station—Zhichunlu
23.08 ha | Community segment
Light, quiet, no disturb
AI as unobtrusive lamp

中段·场段

聚人

Zhichunlu—Qinghua E.
16.8 ha | Phase I
Heartbeat Pt+Grass Theater
Make people walk & stay

北段·线段

点亮

Qinghua E.—N. 5th
30.01 ha | Nature segment
Light, fog, sound flow
护住Children

13 Scenario Cards

01 Heartbeat Train

02 Grass Theater

03 Old Track Whisper

04 Breathing Gate

05 River of Light in Mist

06 Children Sound-Painting

07 Developer Demo Corner

08 Qinghe Wind

09 Community Cinema

10 Pulse Morning Run

11 AI Railway Stories

12 Pulse Data Wall

13 Pillar Light Cascade

Conceptual proposal | Conceptual proposal — All metrics based on provisional boundaries, subject to official survey

Concept: Heartbeat Train



Middle · in-ground lights follow Line 13 trains

Concept: Grass Theater



Middle · sunken lawn by day, stage by night

Concept: Breathing Gate



South · warm white light welcoming home

Concept: River of Light in Mist



North · fiber optics light with footsteps

Concept: Sound-Painting Wall



North · children's voices become colored ripples

Concept: Pillar Light Cascade



North · century of rolling stock on viaduct pillars